Experiment Manual

Analytical Validation of a DIY Structural Vibration Analyzer

Cantilever Steel Bar Natural Frequency Test

Experiment Type:	Free vibration / impact response test
Structure:	Uniform steel cantilever beam
Measured Quantity:	Acceleration versus time
Analysis Method:	Fast Fourier Transform, FFT
Validation Target:	First bending natural frequency
Sensors:	MPU6050 accelerometers, approximately 4.5 g each

Prepared for Structural Vibration Analyzer Validation

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Contents

1 Purpose of the Experiment

The purpose of this experiment is to validate the results obtained from a DIY structural vibration analyzer by comparing experimentally measured natural frequencies with analytically calculated natural frequencies of a steel cantilever beam.

A steel flat bar is fixed at one end and left free at the other end. The beam is excited by either a small initial displacement and release or by a light impact. The accelerometer-based vibration analyzer records the acceleration response of the beam. The recorded time-domain signal is processed using a Fast Fourier Transform, FFT, to identify the dominant vibration frequencies.

The main validation result is the comparison between:

$$f_{\text{analytical}}$$

calculated using Euler-Bernoulli beam vibration theory, and

$$f_{\text{measured}}$$

obtained from the FFT peak measured by the structural vibration analyzer.

Because the MPU6050 sensor has a nonzero mass, the sensor mass is also considered in this manual. Each sensor has an approximate mass of:

$$m_{\text{sensor}} = 4.5\,\text{g} = 0.0045\,\text{kg}$$

This sensor mass may affect the measured natural frequency, especially if the sensor is placed near the free end of the beam. Therefore, this manual includes both the ideal beam-only natural frequency and an approximate corrected natural frequency that includes sensor mass.

The percent error is calculated as:

$$\%\ \text{Error} = \frac{|f_{\text{measured}} - f_{\text{reference}}|}{f_{\text{reference}}} \times 100$$

where $f_{\text{reference}}$ can be either the ideal analytical natural frequency or the corrected analytical natural frequency.

2 Experiment Overview

This experiment uses a uniform steel bar as a simple mechanical structure with predictable natural frequencies. The bar is clamped at one end to create a cantilever beam condition.

The experiment follows this general process:

1. Measure the beam geometry.
2. Determine the beam material properties.
3. Calculate the ideal beam-only natural frequencies.

4. Calculate the beam mass.
5. Estimate whether the 4.5 g sensor mass is significant.
6. Calculate the corrected natural frequency if needed.
7. Mount accelerometer sensors at known positions.
8. Apply known initial conditions to the beam.
9. Record the vibration response.
10. Perform FFT analysis on the measured acceleration signal.
11. Compare analytical and experimental natural frequencies.
12. Calculate percent error.
13. Evaluate possible sources of uncertainty.

3 Required Materials

3.1 Structural Materials

Table 1: Required structural materials

Material	**Description and Purpose**
Steel flat bar	Main test specimen. The beam should have a uniform rectangular cross-section. Recommended free length: 300 mm to 600 mm.
Heavy vise or clamp	Used to create the fixed boundary condition at one end of the beam. The clamp must be as rigid as possible.
Rigid table or workbench	Provides a stable base for the clamp. A heavy workbench is preferred to reduce unwanted motion.
Measuring tape or ruler	Used to measure the free length of the beam.
Digital caliper	Used to measure the width and thickness of the steel bar accurately.
Digital scale	Used to measure the mass of the beam and, if possible, the mass of each sensor and mount.
Small hammer or impact object	Used to apply a repeatable impact excitation to the beam. A rubber-tipped or plastic hammer is preferred.
Optional ruler or displacement marker	Used to apply a consistent initial deflection during pull-and-release testing.

3.2 Instrumentation Materials

Table 2: Required instrumentation materials

Material	Description and Purpose
ESP32 microcontroller	Main data acquisition controller for the structural vibration analyzer.
MPU6050 accelerometer sensors	Sensors used to measure acceleration response of the beam. Each sensor weighs approximately 4.5 g. One to three sensors may be used.
PCA9548A I2C multiplexer	Used if more than one MPU6050 sensor is connected to the ESP32.
Sensor mounting brackets	Used to rigidly attach the sensors to the steel beam. These may be 3D printed, clamped, taped, or bolted depending on the setup. Rigid mounting is strongly preferred.
USB cable	Used to power and communicate with the ESP32.
Computer	Used to record data and process the FFT results.
Python data logger	Records serial data from the ESP32 and saves the data as a CSV file.
FFT analysis script	Processes the CSV data and identifies frequency-domain peaks.

3.3 Recommended Tools

Table 3: Recommended tools

Tool	Purpose
Digital caliper	Accurate measurement of beam cross-section.
Scale	Used to measure the mass of the beam, sensors, and mounts.
Marker	Used to mark sensor locations along the beam.
Tape measure	Used to measure the free beam length.
Level or square	Used to check that the beam is mounted horizontally.
Computer with Python	Used for logging, FFT analysis, and plotting.

4 Beam Geometry and Coordinate System

4.1 Cantilever Beam Configuration

The beam is mounted as a cantilever beam. One end is fixed, and the other end is free.

$$\text{Fixed End} \quad \longrightarrow \quad \text{Free End}$$

The coordinate system used in this experiment is:

- x: distance measured from the fixed end toward the free end.
- y: vertical displacement direction.
- z: lateral direction across the beam width.

The vibration of interest is the vertical bending vibration of the beam. Therefore, the primary measured acceleration direction should be the vertical direction.

4.2 Beam Dimensions

Measure the following dimensions:

Table 4: Measured beam dimensions

Symbol	**Description**	**Unit**
L	Free length of the beam from clamp exit to free end	m
b	Width of the beam	m
h	Thickness of the beam in the bending direction	m
A	Cross-sectional area	m^2
I	Area moment of inertia	m^4
m_{beam}	Mass of the free vibrating portion of the beam	kg

Important: the thickness h must be measured in the direction of bending. If the beam vibrates vertically, h is the vertical thickness of the rectangular cross-section.

5 Theory

5.1 Area and Moment of Inertia

For a rectangular cross-section:

$$A = bh$$

where:

- A is the cross-sectional area,
- b is the beam width,
- h is the beam thickness in the bending direction.

The second moment of area for bending about the neutral axis is:

$$I = \frac{bh^3}{12}$$

The thickness h has a cubic effect on the bending stiffness. Therefore, even small measurement errors in h can noticeably affect the calculated natural frequency.

5.2 Material Properties

For steel, use the following approximate values unless more accurate material data is available:

$$E = 200 \times 10^9 \,\mathrm{Pa}$$

$$\rho = 7850 \,\mathrm{kg/m^3}$$

where:

- E is Young's modulus,
- ρ is material density.

If the exact steel alloy is known, use the manufacturer-provided material properties instead.

5.3 Beam Mass

The mass of the free vibrating portion of the beam is:

$$m_{\mathrm{beam}} = \rho A L$$

where:

- m_{beam} is the mass of the free portion of the beam,
- ρ is material density,
- A is cross-sectional area,
- L is the free length of the beam.

If a scale is available, the beam mass can also be measured directly. However, when using a measured beam mass, make sure the mass corresponds only to the free vibrating portion of the beam.

5.4 Ideal Natural Frequency of a Uniform Cantilever Beam

For a uniform Euler-Bernoulli cantilever beam, the natural frequencies are:

$$f_n = \frac{\lambda_n^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

where:

- f_n is the natural frequency of mode n, in Hz,
- λ_n is the dimensionless mode constant,

- L is the free beam length,
- E is Young's modulus,
- I is the area moment of inertia,
- ρ is density,
- A is cross-sectional area.

The first three cantilever mode constants are:

Table 5: Cantilever beam mode constants

Mode	λ_n
1	1.875
2	4.694
3	7.855

Therefore, the first ideal beam-only natural frequency is:

$$f_{1,\text{ideal}} = \frac{1.875^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

The first mode is the main validation target for this experiment.

6 Sensor Mass Correction

6.1 Why Sensor Mass Matters

Each MPU6050 sensor has an approximate mass of:

$$m_{\text{sensor}} = 4.5\,\text{g} = 0.0045\,\text{kg}$$

This mass is pertinent to the experiment because the sensor is attached directly to the vibrating structure. If a sensor is mounted near the free end of the cantilever beam, it can reduce the measured natural frequency. This happens because the added sensor mass increases the effective vibrating mass of the beam.

The beam-only analytical equation assumes the beam has no added sensor mass. Therefore, the measured frequency may be lower than $f_{1,\text{ideal}}$, even if the vibration analyzer is working correctly.

6.2 Sensor-to-Beam Mass Ratio

To estimate whether the sensor mass is significant, calculate the sensor-to-beam mass ratio:

$$\text{Mass Ratio} = \frac{m_{\text{sensor}}}{m_{\text{beam}}} \times 100$$

where:

$$m_{\text{sensor}} = 0.0045\,\text{kg}$$

and:

$$m_{\text{beam}} = \rho A L$$

Use the following guideline:

Table 6: Sensor mass significance guideline

Sensor Mass Ratio	**Interpretation**
Less than 1%	Usually negligible. The ideal beam-only equation is likely sufficient.
1%–5%	Small but worth mentioning as an uncertainty source.
5%–10%	Significant. Sensor mass should be included in the discussion and correction.
Greater than 10%	Very significant. A corrected analytical model should be used.

6.3 Corrected First Natural Frequency Approximation

For a first-mode approximation, the cantilever beam can be modeled using an equivalent stiffness and an effective modal mass.

The equivalent stiffness at the free end of a cantilever beam is:

$$k_{\text{eq}} = \frac{3EI}{L^3}$$

The effective mass of the first bending mode of a cantilever beam is approximately:

$$m_{\text{eff}} \approx 0.236 m_{\text{beam}}$$

If one sensor is mounted near the free end, the corrected first natural frequency may be approximated as:

$$f_{1,\text{corrected}} \approx \frac{1}{2\pi}\sqrt{\frac{k_{\text{eq}}}{m_{\text{eff}} + m_{\text{sensor}}}}$$

Substituting k_{eq} and m_{eff}:

$$f_{1,\text{corrected}} \approx \frac{1}{2\pi}\sqrt{\frac{3EI/L^3}{0.236 m_{\text{beam}} + m_{\text{sensor}}}}$$

This corrected equation is most appropriate when the main validation sensor is mounted near the free end, such as at:

$$x = 0.90L$$

6.4 Correction for Multiple Sensors

If multiple sensors are mounted along the beam, their effects are not exactly equal. A sensor near the free end affects the first mode more strongly than a sensor near the fixed end.

For a simple first approximation, the sensor at the free end may be considered the most important. The sensors closer to the fixed end can initially be treated as smaller contributors.

If a more conservative correction is desired, the total sensor mass may be estimated as:

$$m_{\text{sensors,total}} = N_{\text{sensors}} m_{\text{sensor}}$$

where N_{sensors} is the number of sensors mounted on the beam.

For three sensors:

$$m_{\text{sensors,total}} = 3(0.0045) = 0.0135\,\text{kg}$$

However, using the full mass of all three sensors in the free-end correction may overestimate the effect, because Sensor 1 and Sensor 2 are not located at the free end. For the first validation, report both:

$$f_{1,\text{ideal}}$$

and

$$f_{1,\text{corrected}}$$

Then discuss whether the measured FFT peak is closer to the ideal or corrected prediction.

7 Sensor Placement

7.1 Recommended Three-Sensor Layout

If using three MPU6050 sensors, place them at the following locations:

Table 7: Recommended sensor positions

Sensor	Position	Coordinate	Purpose
Sensor 1	Near fixed end	$x = 0.10L$	Measures response near the support. Useful for checking base/clamp motion.
Sensor 2	Midspan	$x = 0.50L$	Measures intermediate beam response. Useful for mode shape comparison.
Sensor 3	Near free end	$x = 0.90L$	Main validation sensor. This location should show the strongest first-mode bending response. This sensor has the largest effect on the corrected natural frequency.

If only one sensor is available, place it near the free end:

$$x = 0.90L$$

If two sensors are available, place them at:

$$x_1 = 0.50L$$

$$x_2 = 0.90L$$

7.2 Sensor Axis Orientation

The sensor axis aligned with the vertical bending direction should be used as the primary acceleration signal.

For example, if the beam vibrates vertically and the MPU6050 Z-axis is vertical, then the a_z signal should be used for the FFT validation.

Before testing, write down the sensor orientation:

Table 8: Sensor axis orientation

Sensor Axis	Physical Direction	Used for Analysis?
a_x	____________________	Yes / No
a_y	____________________	Yes / No
a_z	____________________	Yes / No

7.3 Sensor Mounting Requirements

The sensors must be mounted rigidly to the steel beam. Poor mounting can cause the sensor to vibrate independently from the beam, producing incorrect FFT peaks.

Recommended mounting methods, from best to worst:

1. Bolted or screwed rigid sensor bracket.
2. 3D printed bracket clamped firmly to the beam.
3. Thin double-sided adhesive tape.
4. Electrical tape.

Avoid loose wires pulling on the sensor. Wires should be lightly secured to the beam or table with enough slack so they do not restrict beam motion.

8 Initial Conditions

Two excitation methods may be used.

8.1 Method A: Pull-and-Release Free Vibration

This is the preferred method for identifying the first natural frequency.

Initial conditions:

$$y(L, 0) = y_0$$

$$\dot{y}(L, 0) = 0$$

where:

- $y(L, 0)$ is the initial vertical displacement at the free end,
- y_0 is the initial deflection,
- $\dot{y}(L, 0)$ is the initial velocity.

Procedure:

1. Pull the free end of the beam downward by a small known displacement.
2. Hold the beam still.
3. Start recording.
4. Release the beam suddenly without pushing it.
5. Let the beam vibrate freely until the motion decays.

Recommended initial deflection:

$$y_0 = 5\,\mathrm{mm} \text{ to } 20\,\mathrm{mm}$$

The deflection should be small enough that the beam remains in the linear elastic range.

8.2 Method B: Impact Excitation

This method excites multiple frequencies and may help identify higher modes.

Initial conditions are approximately:

$$y(L, 0) = 0$$

$$\dot{y}(L, 0) \neq 0$$

Procedure:

1. Start recording.

2. Tap the beam lightly near the free end.
3. Let the beam vibrate freely.
4. Stop recording after the vibration decays.

The impact should be light and repeatable. Do not hit the sensor directly.

9 Experimental Setup

9.1 Physical Setup

The physical setup should be arranged as follows:

1. Place the steel bar on a rigid workbench.
2. Clamp one end of the steel bar firmly.
3. Measure the free length L from the point where the beam exits the clamp to the free end.
4. Mark sensor locations at $0.10L$, $0.50L$, and $0.90L$.
5. Mount the accelerometers at the marked positions.
6. Connect sensors to the ESP32 or multiplexer.
7. Connect the ESP32 to the computer.
8. Open the serial logger.
9. Confirm that acceleration data is being received before starting the test.

9.2 Recommended Setup Diagram

Include a diagram similar to the following in the final report:

$$\text{Clamp} \quad \boxed{} \begin{bmatrix}\text{Fixed}\\ \text{End}\end{bmatrix} \rule{10cm}{0.4pt} \begin{bmatrix}\text{Free}\\ \text{End}\end{bmatrix}$$

Sensor locations:

$$S_1 = 0.10L$$

$$S_2 = 0.50L$$

$$S_3 = 0.90L$$

The sensor near the free end, S_3, is the primary sensor for natural frequency validation.

10 Data Acquisition Settings

Record the following data acquisition settings before testing:

Table 9: Data acquisition settings

Parameter	Value	
Sampling frequency, f_s	____________________	Hz
Recording duration, T	____________________	s
Number of samples, N	____________________	
Number of sensors	____________________	
Mass of each sensor	0.0045 kg	
Accelerometer range	____________________	g
Gyroscope range, if used	____________________	deg/s
Serial baud rate	____________________	
File name	____________________	

The sampling frequency should satisfy:

$$f_s > 2f_{\max}$$

where $f_{\max}$ is the highest frequency of interest. For better results, use:

$$f_s \geq 10f_1$$

For example, if the expected first natural frequency is 25 Hz, a sampling frequency of at least 250 Hz is recommended.

The FFT frequency resolution is:

$$\Delta f = \frac{f_s}{N}$$

or equivalently:

$$\Delta f = \frac{1}{T}$$

where:

- Δf is the frequency bin spacing,
- f_s is the sampling frequency,
- N is the number of samples,
- T is the recording duration.

A longer recording improves the ability to distinguish nearby frequencies.

11 Pre-Test Checklist

Before running the experiment, complete the following checklist:

Table 10: Pre-test checklist

Task	Completed?
Beam dimensions measured: L, b, and h.	Yes / No
Beam mass estimated or measured.	Yes / No
Sensor mass recorded as approximately 4.5 g each.	Yes / No
Sensor-to-beam mass ratio calculated.	Yes / No
Beam securely clamped to rigid table or vise.	Yes / No
Free length L measured from clamp exit to free end.	Yes / No
Sensor positions marked at desired locations.	Yes / No
Sensors mounted rigidly.	Yes / No
Sensor axes identified.	Yes / No
ESP32 connected to computer.	Yes / No
Serial data confirmed before test.	Yes / No
Sampling frequency recorded.	Yes / No
FFT script ready.	Yes / No

12 Experimental Procedure

12.1 Part 1: Measure Beam Properties

1. Measure the total length of the steel bar.
2. Clamp one end of the beam.
3. Measure the free length L from the clamp exit to the free end.
4. Measure the width b of the beam using a caliper.
5. Measure the thickness h of the beam using a caliper.
6. Record all dimensions in meters.
7. If possible, measure the mass of the free vibrating portion of the beam.
8. Record the mass of each sensor:
$$m_{\text{sensor}} = 0.0045\,\text{kg}$$

12.2 Part 2: Calculate Analytical Natural Frequencies

1. Calculate the cross-sectional area:
$$A = bh$$
2. Calculate the area moment of inertia:
$$I = \frac{bh^3}{12}$$

3. Use steel material properties:

$$E = 200 \times 10^9\,\text{Pa}$$

$$\rho = 7850\,\text{kg/m}^3$$

4. Calculate the beam mass:

$$m_{\text{beam}} = \rho A L$$

5. Calculate the first ideal natural frequency:

$$f_{1,\text{ideal}} = \frac{1.875^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

6. Optionally calculate the second and third natural frequencies:

$$f_2 = \frac{4.694^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

$$f_3 = \frac{7.855^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

12.3 Part 3: Calculate Sensor Mass Effect

1. Calculate the sensor-to-beam mass ratio:

$$\text{Mass Ratio} = \frac{m_{\text{sensor}}}{m_{\text{beam}}} \times 100$$

2. If using one sensor near the free end, use:

$$m_{\text{sensor}} = 0.0045\,\text{kg}$$

3. Calculate the equivalent stiffness:

$$k_{\text{eq}} = \frac{3EI}{L^3}$$

4. Calculate the effective beam mass:

$$m_{\text{eff}} = 0.236 m_{\text{beam}}$$

5. Calculate the corrected first natural frequency:

$$f_{1,\text{corrected}} \approx \frac{1}{2\pi}\sqrt{\frac{k_{\text{eq}}}{m_{\text{eff}} + m_{\text{sensor}}}}$$

6. Compare the corrected frequency with the ideal frequency:

$$f_{1,\text{ideal}} \quad \text{versus} \quad f_{1,\text{corrected}}$$

12.4 Part 4: Mount Sensors

1. Mark the sensor locations using the measured free length L.
2. Mount Sensor 1 at:
$$x = 0.10L$$
3. Mount Sensor 2 at:
$$x = 0.50L$$
4. Mount Sensor 3 at:
$$x = 0.90L$$
5. Ensure all sensors are rigidly attached.
6. Confirm the sensor axis aligned with vertical beam motion.
7. Secure sensor wires so that they do not restrict beam motion.

12.5 Part 5: Run Pull-and-Release Test

1. Open the data logging script.
2. Begin recording acceleration data.
3. Pull the free end of the beam downward by a small displacement, preferably between 5 mm and 20 mm.
4. Hold the beam still for approximately 1 second.
5. Release the beam suddenly without pushing it.
6. Allow the beam to vibrate freely until the motion visibly decays.
7. Stop recording.
8. Save the recorded data file.
9. Repeat the test at least three times.

12.6 Part 6: Run Impact Test

1. Open the data logging script.
2. Begin recording acceleration data.
3. Tap the beam near the free end using a small hammer or plastic object.
4. Do not hit the accelerometer directly.
5. Allow the beam to vibrate freely until the motion decays.
6. Stop recording.
7. Save the recorded data file.
8. Repeat the test at least three times.

13 Data Processing

13.1 Time-Domain Review

Before applying the FFT, inspect the acceleration signal in the time domain.

The expected time-domain response is a decaying oscillation:

$$a(t) \approx A_0 e^{-\zeta \omega_n t} \sin(\omega_d t + \phi)$$

where:

- $a(t)$ is acceleration as a function of time,
- A_0 is the initial acceleration amplitude,
- ζ is damping ratio,
- ω_n is undamped natural frequency,
- ω_d is damped natural frequency,
- ϕ is phase angle.

The signal should show a clear vibration decay after release or impact.

13.2 FFT Processing

The FFT converts the time-domain acceleration signal into the frequency domain.

The frequency peak with the highest amplitude near the analytical prediction is identified as the measured natural frequency:

$$f_{\text{measured}} = \text{dominant FFT peak frequency}$$

Use the sensor closest to the free end, Sensor 3, as the primary validation sensor because it should have the largest first-mode response.

Recommended FFT processing steps:

1. Remove DC offset from acceleration data.
2. Select the primary acceleration axis.
3. Apply a window function if needed.
4. Compute the FFT.
5. Plot amplitude versus frequency.
6. Identify the dominant peak near the analytical first natural frequency.
7. Record the measured peak frequency.

13.3 Validation Calculation

Calculate the percent error relative to the ideal analytical natural frequency:

$$\%\ \mathrm{Error}_{\mathrm{ideal}} = \frac{|f_{\mathrm{measured}} - f_{1,\mathrm{ideal}}|}{f_{1,\mathrm{ideal}}} \times 100$$

Calculate the percent error relative to the corrected analytical natural frequency:

$$\%\ \mathrm{Error}_{\mathrm{corrected}} = \frac{|f_{\mathrm{measured}} - f_{1,\mathrm{corrected}}|}{f_{1,\mathrm{corrected}}} \times 100$$

If the measured value is closer to $f_{1,\mathrm{corrected}}$ than to $f_{1,\mathrm{ideal}}$, then the difference between the beam-only calculation and the experimental result is likely partly caused by the added sensor mass.

Suggested validation criteria:

Table 11: Suggested validation criteria

Percent Error	**Interpretation**
0%–5%	Excellent agreement. Analyzer and setup are performing very well.
5%–10%	Good agreement. Acceptable for a DIY vibration analyzer.
10%–20%	Moderate agreement. Check clamp rigidity, sensor mass, dimensions, and sampling rate.
Greater than 20%	Poor agreement. Experiment setup or signal processing likely needs correction.

14 Experimental Data Tables

14.1 Beam Measurement Table

Table 12: Beam geometry and material properties

Parameter	**Symbol**	**Value**
Free beam length	L	________ m
Beam width	b	________ m
Beam thickness	h	________ m
Cross-sectional area	A	________ m^2
Area moment of inertia	I	________ m^4
Young's modulus	E	________ Pa
Density	ρ	________ $\mathrm{kg/m}^3$
Beam mass	m_{beam}	________ kg
Sensor mass	m_{sensor}	0.0045 kg
Sensor mass ratio	$m_{\mathrm{sensor}}/m_{\mathrm{beam}}$	________ %

14.2 Sensor Position Table

Table 13: Sensor positions

Sensor	Target Position	Measured Position	Primary Axis
Sensor 1	$0.10L$	________ m	______
Sensor 2	$0.50L$	________ m	______
Sensor 3	$0.90L$	________ m	______

14.3 Analytical Frequency Table

Table 14: Analytical natural frequencies

Mode	λ_n	Ideal Analytical Frequency, Hz
1	1.875	__________
2	4.694	__________
3	7.855	__________

14.4 Sensor Mass Correction Table

Table 15: Sensor mass correction calculations

Parameter	Symbol	Value
Beam mass	m_{beam}	________ kg
Sensor mass	m_{sensor}	0.0045 kg
Equivalent stiffness	k_{eq}	________ N/m
Effective modal mass	m_{eff}	________ kg
Corrected first natural frequency	$f_{1,\text{corrected}}$	________ Hz

14.5 Experimental FFT Results Table

Table 16: Experimental FFT results

Trial	Excitation Method	Sensor Used	Measured Peak Frequency, Hz
1	Pull-and-release	Sensor 3	________
2	Pull-and-release	Sensor 3	________
3	Pull-and-release	Sensor 3	________
4	Impact	Sensor 3	________
5	Impact	Sensor 3	________
6	Impact	Sensor 3	________

14.6 Validation Table

Table 17: Validation comparison

Trial	$f_{1,\text{ideal}}$, Hz	$f_{1,\text{corrected}}$, Hz	f_{measured}, Hz	**Best Agreement**
1	________	________	________	Ideal / Corrected
2	________	________	________	Ideal / Corrected
3	________	________	________	Ideal / Corrected
Average	________	________	________	Ideal / Corrected

14.7 Percent Error Table

Table 18: Percent error comparison

Trial	% Error$_{\text{ideal}}$	% Error$_{\text{corrected}}$
1	____________	____________
2	____________	____________
3	____________	____________
Average	____________	____________

15 Expected Results

The expected time-domain result is a decaying sinusoidal acceleration response after the beam is released or impacted.

The expected FFT result is a dominant peak near the analytically calculated first natural frequency f_1.

For the pull-and-release test, the first mode should dominate because the initial deflection shape is similar to the first bending mode.

For the impact test, multiple peaks may appear because the impact can excite higher modes. The first mode should still be visible, especially in the sensor near the free end.

The sensor near the free end should generally show the strongest first-mode response. The sensor near the fixed end may show lower amplitude because the displacement near the fixed support is small.

If the measured first natural frequency is lower than the ideal analytical value but close to the corrected analytical value, the sensor mass is likely influencing the result.

16 Discussion of Error Sources

Possible sources of difference between analytical and experimental results include:

Table 19: Sources of experimental error

Error Source	**Effect on Results**
Clamp flexibility	The theory assumes a perfectly fixed end. If the clamp moves, the measured natural frequency may be lower than the analytical value.
Sensor mass	Each MPU6050 sensor weighs approximately 4.5 g. This added mass can lower the measured natural frequency, especially when mounted near the free end.
Sensor mount mass	Brackets, tape, screws, or adhesive add additional mass to the beam. This should be included if significant.
Beam thickness measurement error	The moment of inertia depends on h^3, so small thickness errors can strongly affect the calculated frequency.
Material property uncertainty	The exact Young's modulus and density may differ from assumed steel values.
Sensor mounting flexibility	A loose mount may create extra vibration peaks or reduce measurement accuracy.
Wire stiffness	Wires can add damping or stiffness if they restrict beam motion.
Sampling frequency limitations	Low sampling rate can cause inaccurate frequency measurement or aliasing.
FFT resolution	Short recording duration gives larger frequency spacing and less accurate peak identification.
Nonlinear excitation	Large deflection can make the beam response nonlinear and reduce agreement with linear theory.

17 Acceptance Criteria

The experiment validates the structural vibration analyzer if the following conditions are met:

1. The time-domain signal shows a clear decaying vibration response.
2. The FFT plot shows a clear dominant peak.
3. The dominant peak is near the calculated first natural frequency.
4. The measured peak is compared against both the ideal and corrected natural frequency.
5. The percent error is within an acceptable range.
6. Repeated trials produce similar measured frequencies.

For this experiment, the recommended acceptance criterion is:

$$\% \text{ Error} \leq 10\%$$

The corrected frequency comparison should be emphasized if the sensor mass ratio is greater than approximately 5%.

18 Minimum Report Requirements

The final experiment report should include:

1. Purpose of the experiment.
2. Photograph or diagram of the setup.
3. Beam dimensions and material properties.
4. Sensor mass and sensor-to-beam mass ratio.
5. Sensor positions and sensor axis orientation.
6. Ideal analytical natural frequency calculation.
7. Corrected natural frequency calculation including sensor mass.
8. Time-domain acceleration plots.
9. FFT plots.
10. Measured natural frequency values.
11. Percent error calculation using ideal frequency.
12. Percent error calculation using corrected frequency.
13. Discussion of uncertainty and possible error sources.
14. Conclusion on whether the analyzer was validated.

19 Example Calculation Template

Use this section as a calculation template.

Given:

$$L = \underline{\hspace{3cm}}\ \mathrm{m}$$

$$b = \underline{\hspace{3cm}}\ \mathrm{m}$$

$$h = \underline{\hspace{3cm}}\ \mathrm{m}$$

$$E = 200 \times 10^9\,\mathrm{Pa}$$

$$\rho = 7850\,\mathrm{kg/m^3}$$

$$m_{\text{sensor}} = 0.0045\,\mathrm{kg}$$

Calculate area:

$$A = bh$$

Calculate second moment of area:

$$I = \frac{bh^3}{12}$$

Calculate beam mass:

$$m_{\text{beam}} = \rho A L$$

Calculate ideal first natural frequency:

$$f_{1,\text{ideal}} = \frac{1.875^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

Calculate equivalent stiffness:

$$k_{\text{eq}} = \frac{3EI}{L^3}$$

Calculate effective modal mass:

$$m_{\text{eff}} = 0.236 m_{\text{beam}}$$

Calculate corrected first natural frequency:

$$f_{1,\text{corrected}} \approx \frac{1}{2\pi}\sqrt{\frac{k_{\text{eq}}}{m_{\text{eff}} + m_{\text{sensor}}}}$$

Measured frequency from FFT:

$$f_{\text{measured}} = \underline{\hspace{4cm}}\ \text{Hz}$$

Percent error using ideal frequency:

$$\%\ \text{Error}_{\text{ideal}} = \frac{|f_{\text{measured}} - f_{1,\text{ideal}}|}{f_{1,\text{ideal}}} \times 100$$

Percent error using corrected frequency:

$$\%\ \text{Error}_{\text{corrected}} = \frac{|f_{\text{measured}} - f_{1,\text{corrected}}|}{f_{1,\text{corrected}}} \times 100$$

20 Conclusion Statement Template

The following conclusion statement may be used in the final report:

> The structural vibration analyzer was tested using a steel cantilever beam with known geometric and material properties. The first bending natural frequency was calculated analytically using Euler-Bernoulli beam theory. Because the MPU6050 sensor has an approximate mass of 4.5 g, a corrected first natural frequency was also calculated using an equivalent stiffness and effective modal mass approximation. The beam was excited using a free-vibration pull-and-release method and an impact method. Acceleration data was recorded using MPU6050 sensors and processed using FFT analysis. The dominant measured frequency was compared with both the ideal analytical prediction and the corrected prediction. The resulting percent error was __________% relative to the ideal value and __________% relative to the corrected value. This indicates that the analyzer _______________ the first natural frequency of the test structure with acceptable accuracy.

21 Appendix A: Quick Procedure Summary

1. Clamp steel bar as a cantilever.
2. Measure L, b, and h.
3. Calculate $A = bh$.
4. Calculate $I = bh^3/12$.
5. Calculate $m_{\text{beam}} = \rho AL$.
6. Record sensor mass as $m_{\text{sensor}} = 0.0045\,\text{kg}$.
7. Calculate $f_{1,\text{ideal}}$.
8. Calculate $f_{1,\text{corrected}}$, if sensor mass is significant.
9. Mount Sensor 3 near $0.90L$.
10. Record acceleration during pull-and-release vibration.
11. Run FFT analysis.
12. Identify measured frequency peak.
13. Calculate percent error against both ideal and corrected frequency.
14. Repeat for at least three trials.

22 Appendix B: Recommended File Naming

Use clear file names for data organization:

Table 20: Recommended file naming structure

File Type	Recommended Name
Raw data, trial 1	cantilever_trial1_raw.csv
Raw data, trial 2	cantilever_trial2_raw.csv
Raw data, trial 3	cantilever_trial3_raw.csv
FFT results, trial 1	cantilever_trial1_fft.csv
Time-domain plot	cantilever_time_domain.png
FFT plot	cantilever_fft_spectrum.png
Final report	cantilever_validation_report.pdf

23 Appendix C: Important Notes

- The first bending mode is the easiest and most reliable validation target.
- Do not use large deflections because the analytical equation assumes linear vibration.
- The clamp must be as rigid as possible.
- The free length must be measured from the actual fixed boundary, not from the physical end of the clamp.
- The sensor near the free end should be used as the main validation sensor.
- Each MPU6050 sensor weighs approximately 4.5 g.
- The sensor mass may lower the measured natural frequency.
- The FFT peak may not be exactly equal to the ideal analytical value due to sensor mass, clamp flexibility, and material uncertainty.
- If the measured frequency is consistently lower than theory, check clamp flexibility and added sensor mass.
- If the measured value is closer to the corrected frequency than the ideal frequency, sensor mass is likely affecting the experiment.
- If the FFT plot contains many unexpected peaks, check for loose sensor mounting, wire motion, or external vibration.